Day 8 - Practice Question
Week-2_Day 1(Day-8): React Basics & Component Fundamentals

Title: “BookVerse: Welcome Page and Featured Books Section”
User Story
As a visitor,
I want to view a list of featured books on the home page,
So that I can get an overview of popular books available.

Acceptance Criteria

- Initialize a React app using create-react-app.
- Create a BookCard functional component displaying book title, author,and price.
- Create a BookList parent component that renders multiple BookCardcomponents using props.
- Use state to manage and toggle between two view modes – “Grid” and“List”.
- Handle button click events to switch layout.
- Demonstrate controlled components with a simple search input box that filters the displayed books.

technical Focus
React overview, JSX, Virtual DOM
Functional Components, Props, and State
Event Handling, Controlled Components
Time Estimate: 60 mins

Solution:

First we need to create a react app
Use:npx create-react-app bookverse

```
cd bookverse
npm install
npm start
```

Structure of the codes:

```
Bookverse/src/
├── components/
│   ├── BookCard.js
│   └── BookList.js
├── App.js
├── App.css
└── index.js
```

Codes:
 src/components/BookCard.js:

```
import React from "react";

function BookCard({ title, author, price, viewMode }) {
  return (
    <div className={viewMode === "grid" ? "book-card-grid" : "book-card-
list"}>
      <h3>{title}</h3>
      <p><strong>Author:</strong> {author}</p>
      <p><strong>Price:</strong> ₹{price}</p>
    </div>
  );
}

export default BookCard;
```

2. src/components/BookList.js:

```
import React, { useState } from "react";
import BookCard from "./BookCard";

function BookList() {

  const booksData = [
    { id: 1, title: "The Alchemist", author: "Paulo Coelho", price: 399 },
    { id: 2, title: "Atomic Habits", author: "James Clear", price: 499 },
    { id: 3, title: "Ikigai", author: "Hector Garcia", price: 299 },
    { id: 4, title: "Rich Dad Poor Dad", author: "Robert Kiyosaki", price: 450 },
    { id: 5, title: "Wings of Fire", author: "A.P.J Abdul Kalam", price: 350 }
  ];

  const [viewMode, setViewMode] = useState("grid");
  const [searchTerm, setSearchTerm] = useState("");

  const filteredBooks = booksData.filter(book =>
    book.title.toLowerCase().includes(searchTerm.toLowerCase())
  );
```

```
  return (
    <div>
      <h2>Featured Books</h2>

      <input
        type="text"
        placeholder="Search books..."
        value={searchTerm}
        onChange={(e) => setSearchTerm(e.target.value)}
      />

      <div style={{ marginTop: "10px" }}>
        <button onClick={() => setViewMode("grid")}>Grid View</button>
        <button onClick={() => setViewMode("list")}>List View</button>
      </div>

      <div className={viewMode === "grid" ? "grid-container" : "list-container"}>
        {filteredBooks.map(book => (
          <BookCard
            key={book.id}
            title={book.title}
            author={book.author}
            price={book.price}
            viewMode={viewMode}
          />
        ))}
      </div>
    </div>
  );
}

export default BookList;
```

3. src/App.js:

```
import React from "react";
import BookList from "./components/BookList";
import "./App.css";

function App() {
  return (
    <div className="App">
      <h1>BookVerse</h1>
```

```
      <BookList />
    </div>
  );
}

export default App;
```

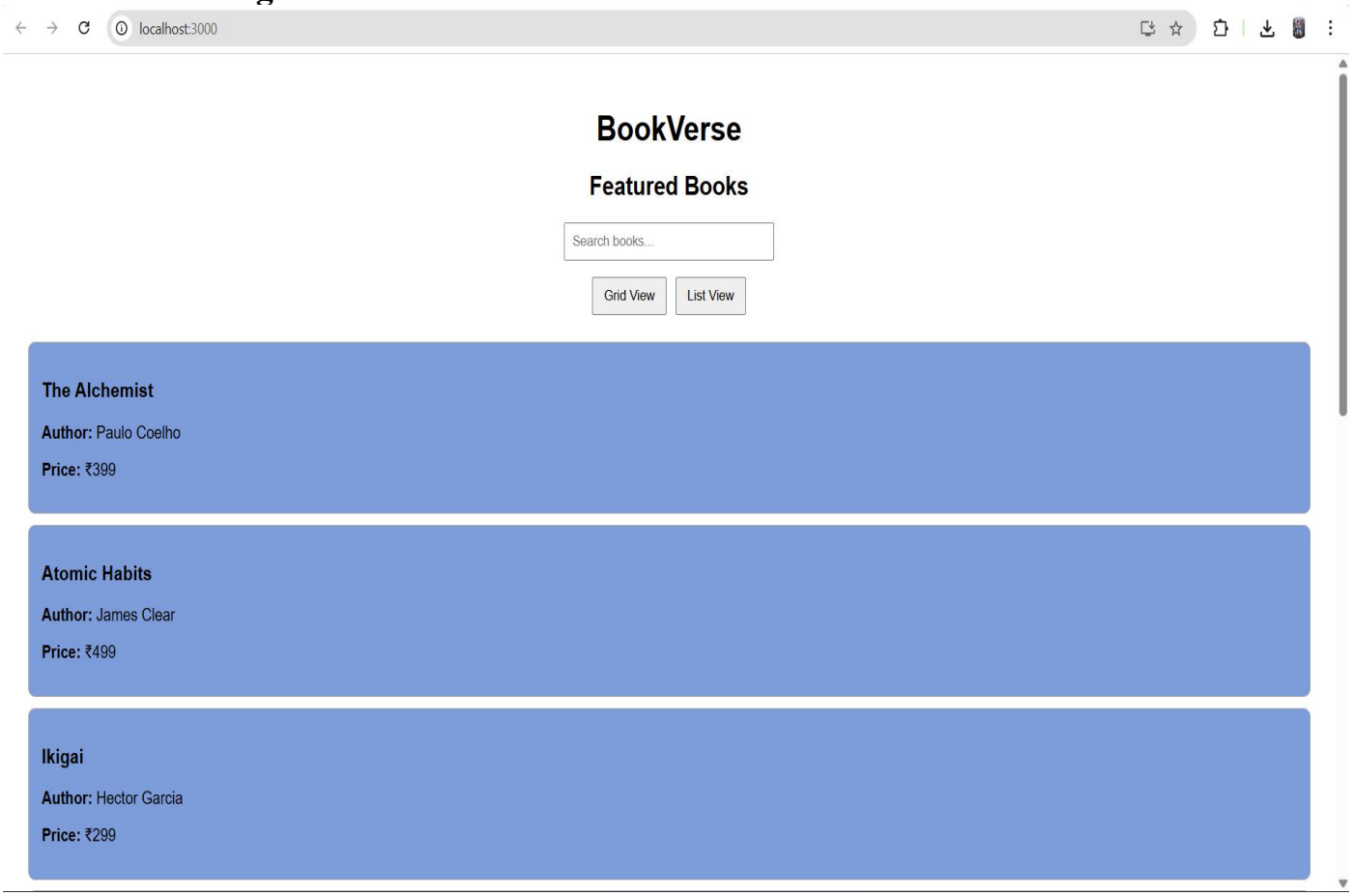
4. src/App.css:

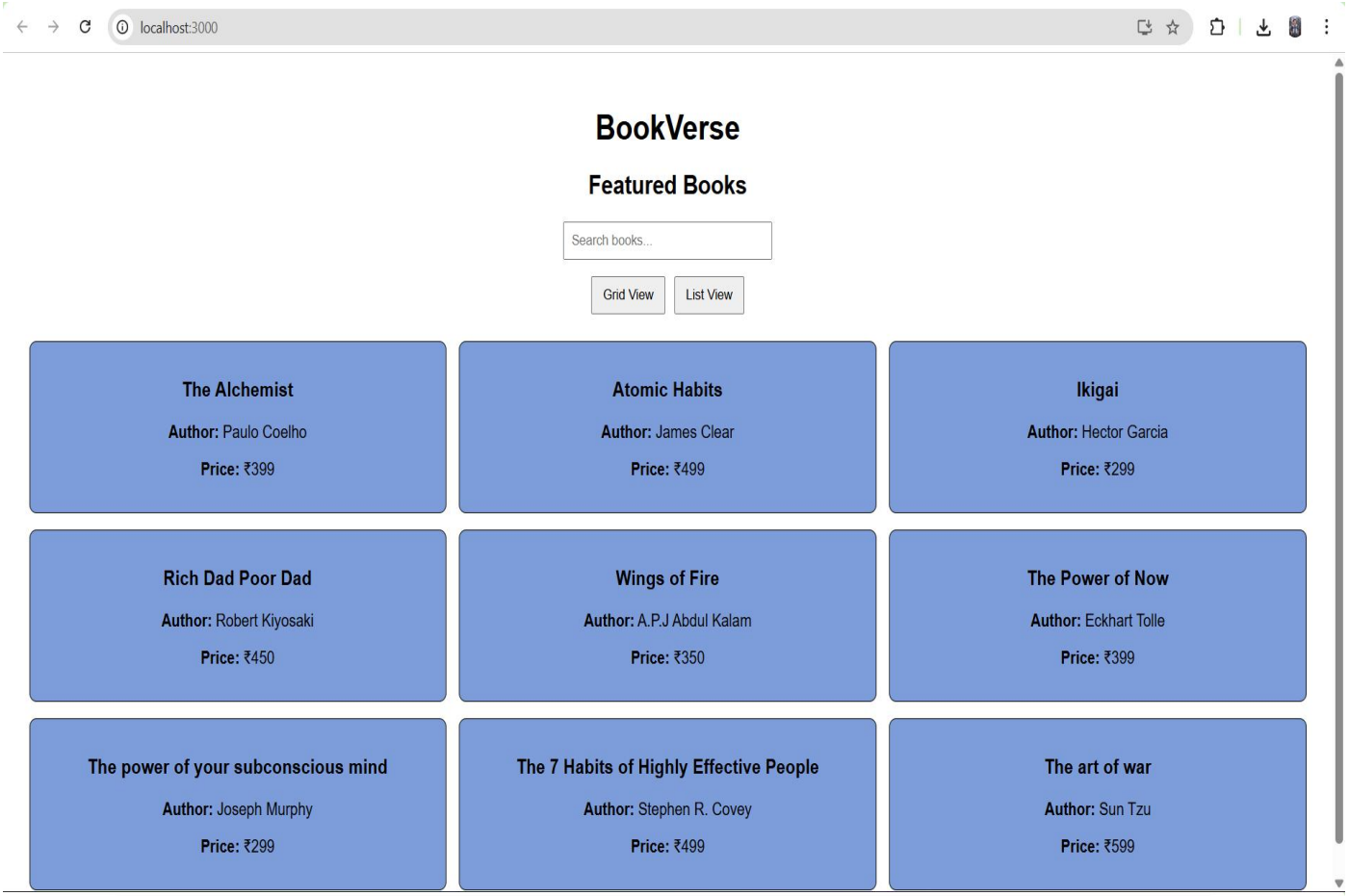
```
.App {
  text-align: center;
  padding: 20px;
  font-family: Arial, Helvetica, sans-serif;
}

.grid-container {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 15px;
  margin-top: 20px;
}

.list-container {
  display: flex;
  flex-direction: column;
  gap: 10px;
  margin-top: 20px;
}

.book-card-grid {
  border: 1px solid #ccc;
  padding: 15px;
  border-radius: 8px;
  background-color: #f5f5f5;
}

.book-card-list {
  border: 1px solid #aaa;
  padding: 15px;
  border-radius: 8px;
  background-color: #e8f0ff;
  text-align: left;
}
```

```
button {
  margin: 5px;
  padding: 8px 12px;
  cursor: pointer;
}

input {
  padding: 8px;
  width: 220px;
}
```

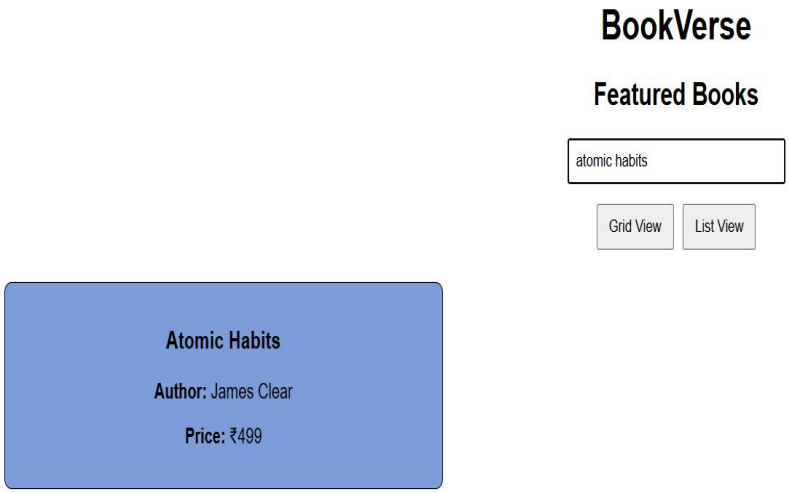
To run use npm start,and then use this link to host it in web-browser:http://localhost:3000

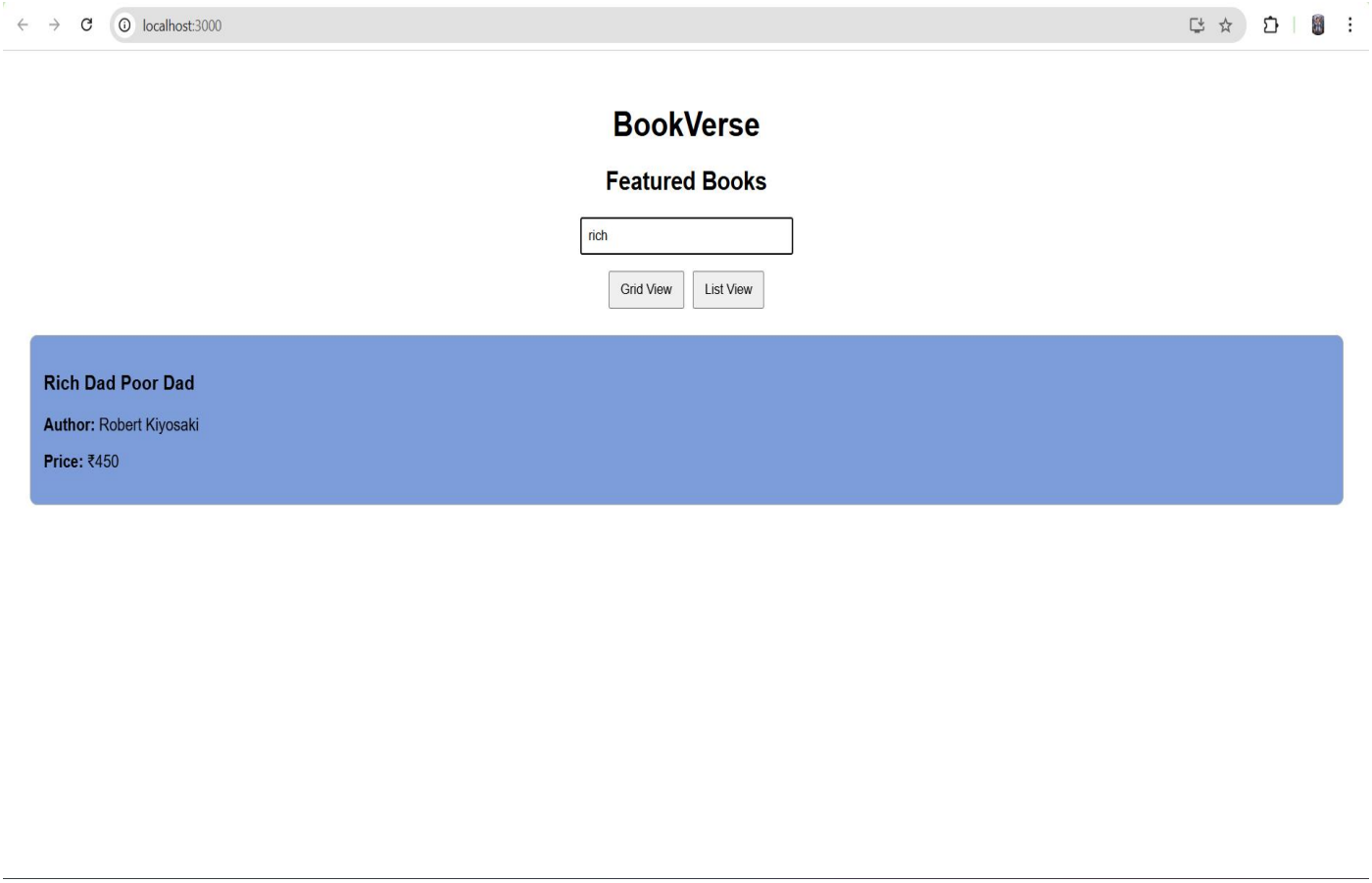
WEBPAGE/images:




localhost:3000
BookVerse
Featured Books
Search books...
Grid View
List View
The Alchemist
Author: Paulo Coelho
Price: ₹399
Atomic Habits
Author: James Clear
Price: ₹499
Ikigai
Author: Hector Garcia
Price: ₹299
Rich Dad Poor Dad
Author: Robert Kiyosaki
Price: ₹450
Wings of Fire
Author: A.P.J Abdul Kalam
Price: ₹350
The Power of Now
Author: Eckhart Tolle
Price: ₹399
The power of your subconscious mind
Author: Joseph Murphy
Price: ₹299
The 7 Habits of Highly Effective People
Author: Stephen R. Covey
Price: ₹499
The art of war
Author: Sun Tzu
Price: ₹599

localhost:3000


BookVerse
Featured Books
atomic habits
Grid View
List View
Atomic Habits
Author: James Clear
Price: ₹499


localhost:3000
BookVerse
Featured Books
rich
Grid View
List View
Rich Dad Poor Dad
Author: Robert Kiyosaki
Price: ₹450